

Software Requirements Specification

GraduPredict Graduate Placement Prediction System

BS Computer Science Final Year Project
Version 1.0 | October 2026

Document Purpose

This document defines the software requirements for a web application that estimates graduate placement likelihood from academic and employability information. It gives a single student a bounded, testable implementation scope and gives supervisors a basis for acceptance testing.

1 Introduction

1.1 Product Scope

GraduPredict accepts a student profile, validates it, calculates a placement probability with Logistic Regression, assigns an outcome and risk category, stores approved results, and provides dashboard, history, export, and model-information views.

1.2 Intended Users

User	Responsibilities
Student or counselor	Enter valid profile data, review the estimate, and save appropriate records.
Project administrator	Train the model, maintain data and model files, and verify database and application operation.
Supervisor or examiner	Evaluate requirements, design, model workflow, testing evidence, and viva demonstration.

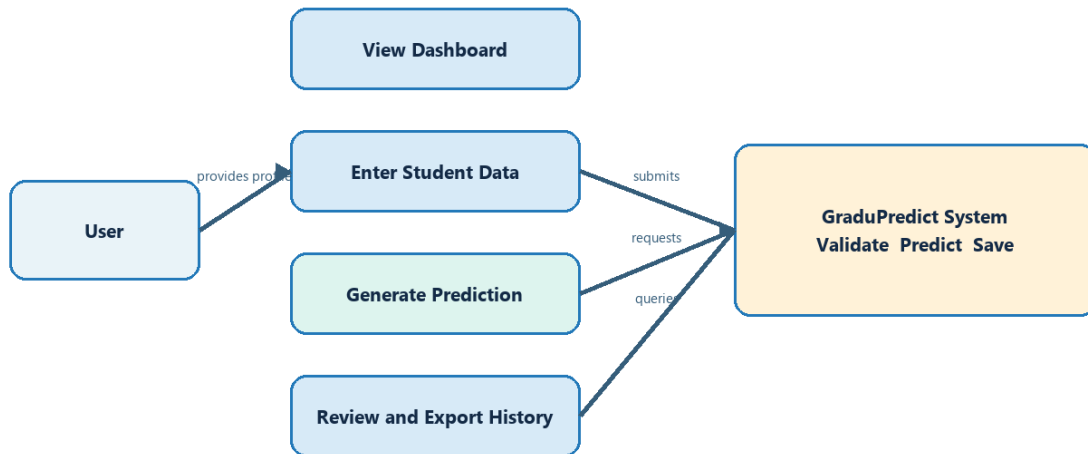
1.3 Constraints and Assumptions

The implementation is designed for one developer and local or Streamlit Community Cloud deployment. It uses a binary target. The included dataset is for demonstration; final reported research results must come from the selected and documented Kaggle dataset. The estimate supports counseling and does not make hiring decisions.

2 Overall Description

The system is a standalone Streamlit application. Prediction parameters are read from a version-portable JSON file. Saved results are stored in SQLite. The interface has four pages: Dashboard, New Prediction, Prediction History, and About Model.

GraduPredict Use Case Overview



3 Functional Requirements

ID	Name	Requirement	Acceptance test
FR 01	Navigation	The system shall display Dashboard, New Prediction, Prediction History, and About Model pages.	All four pages open without errors.
FR 02	Data entry	The system shall capture student ID and eleven numeric or binary model inputs.	All fields accept values within their defined ranges.
FR 03	Validation	The system shall reject missing student IDs and values outside interface limits.	An explanatory message appears and prediction is not saved.
FR 04	Prediction	The system shall calculate a probability from the stored Logistic Regression parameters.	A probability from 0 to 1 is returned.
FR 05	Classification	The system shall label the result Likely placed or At risk and assign Low, Medium, or High risk.	Labels agree with configured thresholds.
FR 06	Result display	The system shall show outcome, probability, risk, and a responsible-use note.	All result elements appear after a valid request.
FR 07	Persistence	The system shall save an accepted result to SQLite.	A row with student ID, result, probability, risk, and timestamp is created.
FR 08	History	The system shall display saved results in descending order.	Newest record appears first.
FR 09	Filtering	The system shall filter history by outcome.	Only matching rows remain.
FR 10	Export	The system shall export the	Downloaded CSV contains

		displayed history as CSV.	displayed columns and rows.
FR 11	Dashboard	The system shall summarize counts, recent records, and model accuracy.	Metrics update from saved data and model metadata.
FR 12	Model information	The system shall present the algorithm, target, metrics, data note, and limitation.	The About Model page contains every item.

4 Data Requirements

Field	Type	Rule
ssc_percentage	Numeric	Required model feature
hsc_percentage	Numeric	Required model feature
degree_percentage	Numeric	Required model feature
employability_score	Numeric	Required model feature
internships	Numeric	Required model feature
projects	Numeric	Required model feature
communication_rating	Numeric	Required model feature
technical_rating	Numeric	Required model feature
backlogs	Numeric	Required model feature
work_experience	Numeric	Required model feature
training_completed	Numeric	Required model feature
student_id	Text	Required and nonblank
placed	Binary	Training target only

5 Nonfunctional Requirements

ID	Area	Requirement
NFR 01	Usability	A new user shall complete a prediction without training within three minutes.
NFR 02	Performance	A local prediction shall normally appear within two seconds after submission.
NFR 03	Portability	The project shall run on Windows, Linux, Kaggle notebooks, and Streamlit hosting with Python 3.10 or later.
NFR 04	Compatibility	Model parameters shall use JSON rather than a version-sensitive pickle.
NFR 05	Privacy	The system shall avoid names, contact details, and protected hiring decisions.
NFR 06	Maintainability	Prediction, interface, training, and persistence responsibilities shall be separated.
NFR 07	Reliability	The database shall be initialized automatically if it is missing.
NFR 08	Reproducibility	Training shall use a documented random seed and saved feature order.

6 Acceptance Criteria

The project is accepted when the model can be trained, the application starts, valid input produces a result, invalid input is rejected, a result can be saved, history can be filtered and exported, and the supplied automated prediction tests pass.

7 Requirements Traceability

Requirement group	Implementation	Verification
FR 01 to FR 06	app.py and prediction.py	Manual interface demonstration and unit tests
FR 07 to FR 10	app.py and gradupredict.db	Database record and CSV inspection
FR 11 to FR 12	app.py and placement_model.json	Page inspection
NFR 03 to NFR 08	requirements, JSON model, README, training script	Clean setup and file review